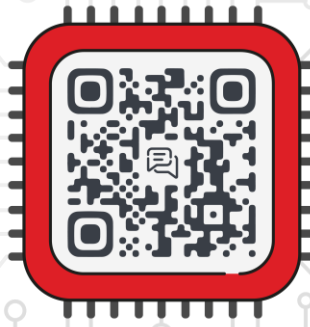




ملحق

كلية الهندسة المعلوماتية

السنة الثالثة

ملحق امتحاني



محتوى مجاني غير مخصص للبيع التجاري

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أساسيات الشبكات

السلام عليكم يا أصدقاء

يتضمن هذا الملحق أسئلة اختيار من متعدد مستخلصة من المحاضرات ومن بعض المواقع ومن المرجع ومع ذلك، لا يُغني هذا عن ضرورة دراسة المحاضرات قبل البدء بحل هذه التمارين، مع العلم إنه عمل طلابي.
ملاحظة : اضغط على اسم المحاضرة للانتقال على القسم الخاص بها



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Network Model

1) What is the main **advantage** of using layered tasks in **communication systems**?

A. It makes the system more complex	C. It eliminates the need for protocols.	B
B. It allows independent development and easier troubleshooting	D. It reduces the number of devices needed in communication	

troubleshooting = استكشاف الأخطاء وإصلاحها

2) In the analogy of sending a letter, what does the post office represent in a network model?

A. The sender	C. The transmission medium and intermediate nodes	C
B. The receiver	D. The application layer	

3) Which organization developed the OSI model?

A. IEEE	C. IETF	B
B. ISO	D. ITU	

4) How many layers does the OSI model have?

A. 5	C. 7	C
B. 6	D. 8	

ليس المهم عدد الأخطاء التي ترتكبها ولا بطء التقدم
الذي تحرزه فأنت لا تزال متقدماً على كل من لا يحاول



5) What is **encapsulation** in the OSI model?

A- The process of encrypting data for security	C- The process of compressing data to reduce size	B
B- The process of adding headers and trailers as data moves down the layers	D- The process of translating data between different formats	

6) What happens when data is sent from one device to another in the OSI model?

A- The data is transmitted directly without modification	C- The OSI model compresses the data before sending it	B
B- The data passes through the OSI layers, where headers are added and removed accordingly	D- The data is encrypted and sent directly to the application layer of the receiver	

7) What does the acronym **ISO** stand for?

A. International System Organization	C. International Organization for Standardization	B
B. International Standards Organization	D. Internet Security Organization	

توضيح:

اختصار **ISO** في المرجع هو **International Standards Organization** لكن لا يوجد منظمة رسمية بـ هذا الاسم
اسم المنظمة الرسمي هو **International Organization for Standardization**

8) What is the **main role** of the International Organization for Standardization (ISO)?

A. To regulate internet services globally	C. To enforce cybersecurity laws worldwide	B
B. To develop and publish international standards across various industries	D. To create networking hardware	



9) What does **OSI** stand for in networking?

A. Open Source Interconnection	C. Operational Security Interface	B
B. Open Systems Interconnection	D. Organizational System Integration	

10) What is the main difference between **ISO** and **OSI**?

A. ISO is a networking protocol, while OSI is a global organization	C. OSI and ISO are two different networking protocols	B
B. ISO is a standardization organization, while OSI is a networking model developed by ISO	D. ISO is only used for internet communication, while OSI is for local networks	

11) Which of the following statements about **ISO** and **OSI** is **FALSE**?

A. ISO developed the OSI model as a global networking standard	C. OSI is a conceptual model, not a protocol suite	B
B. OSI is a practical implementation used in all modern networks	D. ISO is responsible for various international standards, not just networking	

ملاحظة :

نظام **OSI** هو مجرد نموذج نظري لم يتم اعتماده بشكل عملي.

12) Which of the following statements about **OSI** and **TCP/IP** is **FALSE**?

A. OSI is a conceptual model, while TCP/IP is a practical implementation	C. OSI is used more in modern networks than TCP/IP	C
B. OSI has 7 layers, while TCP/IP has either 4 or 5 layers	D. TCP/IP was developed before the OSI model	

13) What is the primary function of the **Physical Layer**?

a) Routing data between devices	c) Ensuring data integrity	B
b) Transmitting bits over a physical medium	d) Encrypting data for security	

14) Which of the following **devices** operates at the **Physical Layer**?

a) Router	c) Hub	C
b) Switch	d) Firewall	

15) Which of the following is **NOT** a responsibility of the **Physical Layer** ?

a) Defining the type of cable	c) Determining the data rate	B
b) Assigning IP addresses	d) Converting bits into signals	

16) Which of the following is an **example** of a **physical transmission medium**?

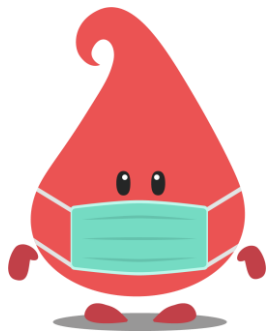
a) HTTP protocol	c) MAC address	B
b) Fiber optic cable	d) TCP protocol	

17) What is the unit of data handled by the **Physical Layer**?

a) Packet	c) Bit	C
b) Frame	d) Segment	

18) Which of the following is the correct order of the OSI model layers from **top to bottom**?

A. Transport, Network, Application, Session, Data Link, Physical, Presentation	C. Physical, Data Link, Network, Transport, Session, Presentation, Application	B
B. Application, Presentation, Session, Transport, Network, Data Link, Physical	D. Network, Transport, Data Link, Physical, Application, Presentation, Session	



19) In the OSI model, **how** do layers interact with each other?

A. Each layer interacts with all other layers	C. The top three layers interact, while the bottom four layers remain independent	B
B. Each layer only interacts with the layer directly above and below it	D. Layers communicate in a random order	

20) What happens to data as it moves down the OSI model layers on the sender's side?

A. It remains unchanged until it reaches the physical layer	C. The data is converted into an IP address at each layer	B
B. Each layer adds its own header (Encapsulation)	D. The application layer encrypts the data permanently	

21) What is the type of relationship between the Transport Layer on the sender's side and the Transport Layer on the receiver's side in the OSI model ?

A. Hierarchical Relationship	C. Cross-Layer Communication	B
B. Peer-to-Peer Relationship	D. Encapsulation Relationship	

22) What is the purpose of the OSI model?

A. To define a standard framework for network communication between different systems	C. To ensure that only identical devices can communicate	A
B. To replace all other networking models	D. To enforce internet regulations	

23) What does the Physical Layer use to synchronize communication between devices?

a) IP addresses	c) MAC addresses	B
b) Clocks and timing signals	d) Error detection codes	

24) What is the purpose of the Physical Layer's transmission mode?

a) To determine the direction of data flow	c) To assign IP addresses	A
b) To encrypt data for security	d) To manage network congestion	

25) Identify the item that does **NOT** belong to the physical media used in the Physical Layer:

a) Fiber optic cable	c) UTP cable	D
b) Wireless signals	d) IP address	

26) Which transmission mode allows devices to send and receive data at the same time?

A. Simplex	C. Full-Duplex	C
B. Half-Duplex	D. Serial	

27) Which of the following is **NOT** a function of the Physical Layer?

A. Defining the physical medium for data transmission	C. Determining the best path for data packets	C
B. Converting data into signals for transmission	D. Establishing data transmission rates	

28) What are the two **main types of signals** used in the **Physical Layer**?

A. Digital and Analog	C. Logical and Physical	A
B. Encoded and Decoded	D. Unicast and Multicast	

29) The **Physical Layer** is responsible for moving bits between:

A. The application layer and transport layer	C. One hop (node) to the next in the network	C
B. The sender and the receiver directly	D. The network layer and data link layer	

30) What is **NOT** a characteristic of the **Physical Layer**?

A. It defines the hardware specifications for network transmission	C. It establishes how signals are transmitted through cables or wireless media	B
B. It manages logical addressing for devices	D. It controls the synchronization of signals between sender and receiver	

31) How does the **Data Link Layer** handle **errors** in data transmission?

A. By retransmitting corrupted frames	C. By choosing the shortest path for data transmission	A
B. By encrypting the data before sending it	D. By assigning a new IP address to the frame	



32) What is a key **difference** between the Physical Layer and the Data Link Layer?

A. The Physical Layer deals with frames, while the Data Link Layer deals with bits	C. The Data Link Layer transmits electrical signals, while the Physical Layer determines MAC addresses	B
B. The Physical Layer transmits raw bits, while the Data Link Layer organizes bits into frames	D. Both layers perform the same functions	

33) What is the primary function of the **Data Link Layer** in the OSI model?

A. Routing packets between different networks	C. Transmitting frames within the same network	C
B. Moving bits from one device to another	D. Encrypting data before transmission	

34) What is the unit of data at the **Data Link Layer**?

A. Bit	C. Frame	C
B. Packet	D. Segment	

35) What type of addressing is used in the **Data Link Layer**?

A. IP Address	C. Port Number	B
B. MAC Address	D. Subnet Mask	

36) What mechanism does the **Data Link Layer** use for **error detection**?

A. IP Routing	C. DHCP	B
B. CRC (Cyclic Redundancy Check)	D. DNS	

37) What is the main purpose of **flow control** in the **Data Link Layer**?

a) To encrypt data for security	c) To assign IP addresses	B
b) To ensure the sender does not overwhelm the receiver	d) To detect errors in transmitted data	

38) Which of the following is a function of **error control** in the **Data Link Layer**?

a) To assign IP addresses	c) To encrypt data for security	D
b) To detect the errors in transmitted frames	d) To detect and correct errors in transmitted frames	

ملاحظة:

طبقة **Data Link** تقوم بـ إيجاد الخطأ وإصلاحه وذلك بـ إعادة إرساله.

39) Which of the following functions belongs to the **Data Link Layer**?

a) Routing data between networks	c) Encrypting data for secure communication	B
b) Detecting and correcting errors in transmitted frames	d) Establishing, managing, and terminating sessions	

40) Which of the following functions does **NOT** belong to the **Data Link Layer**?

a) Framing	c) Flow control	D
b) Physical addressing (MAC addresses)	d) Translating data into a readable format	

41) Which network **devices** operate at the **Data Link Layer**?

A. Routers	C. Hubs	B
B. Switches and Bridges	D. Firewalls	

42) What happens to the MAC address when a frame is forwarded to another network?

A. It remains unchanged	C. It is converted into an IP address	B
B. It is replaced with a new MAC address for the next hop	D. It is encrypted for security	

43) When a frame reaches its destination device, what does the Data Link Layer do?

A. It checks the MAC address and removes the Data Link header	C. It forwards the frame to the next router	A
B. It converts the frame into an IP packet	D. It encrypts the frame for security	

44) What type of delivery occurs at the Physical Layer ?

A. Hop-to-Hop	C. Both Hop-to-Hop and Source-to-Destination	A
B. Source-to-Destination	D. Neither	

45) The Data Link Layer primarily handles which type of delivery?

A. Hop-to-Hop	C. Network-to-Network	A
B. Source-to-Destination	D. Process-to-Process	

46) What type of delivery is primarily handled by the Network Layer?

A. Hop-to-Hop	C. Physical-to-Physical	B
B. Source-to-Destination	D. None of the above	

47) What type of network primarily operates at the **Data Link Layer** of the OSI model?

A. Wide Area Network (WAN)	C. Metropolitan Area Network (MAN)	B
B. Local Area Network (LAN)	D. Personal Area Network (PAN)	

48) What type of network does the **Network Layer** primarily operate in?

A. Local Area Network (LAN)	C. Storage Area Network (SAN)	B
B. Wide Area Network (WAN)	D. Virtual Private Network (VPN)	

49) Which layer is the only one that **adds** both a **header** and a **trailer** during the encapsulation process?

a) Physical Layer	c) Network Layer	B
b) Data Link Layer	d) Transport Layer	

50) What information is typically **found in the header** of a **Data Link Layer frame**?

a) Source and destination IP addresses	c) Source and destination port numbers	B
b) Source and destination MAC addresses	d) Sequence and acknowledgment numbers	

51) What is the primary purpose of the trailer in a Data Link Layer frame?

a) To store source and destination MAC addresses	c) To include sequence numbers for data reassembly	B
b) To provide error detection information	d) To encrypt the data for security	

52) What type of delivery is primarily handled by the **Transport Layer**?

a) Hop-to-Hop	c) Source-to-Destination	B
b) Process-to-Process	d) Bit-by-Bit	

53) What is the role of **Synchronization** in the **Physical Layer**?

A. To encrypt data before transmission	C. To define logical addressing for devices	B
B. To keep the sender and receiver in sync during data transmission	D. To determine the shortest route for data packets	

54) How is data transmission **speed measured** in the **Physical Layer**?

A. Bits per second (bps)	C. Packets per second (pps)	A
B. Frames per second (fps)	D. Segments per second (sps)	

55) In a **shared network medium**, what is the role of Access Control at the **Data Link Layer** ?

A. It determines the best route for data transmission across networks.	C. It encrypts data before sending it over the network.	B
B. It ensures that only one device transmits at a time to avoid collisions.	D. It converts MAC addresses into IP addresses.	

56) What is the primary function of the **Network Layer** in the OSI model?

A. Moving frames within a local network	C. Managing encryption and data compression	B
B. Delivering packets from source to destination across multiple networks	D. Controlling access to the transmission medium	

57) What type of addressing is used in the Network Layer?

A. MAC Address	C. Port Number	B
B. IP Address	D. Physical Address	

58) What is the main function of a router in the Network Layer?

A. Forwarding packets between different networks	C. Assigning MAC addresses to devices	A
B. Moving frames within a local network	D. Encrypting network data	

59) Which of the following is **NOT** a function of a **router** in the **Network Layer** ?

A. Logical addressing using IP addresses	C. Forwarding packets to their destination based on IP addresses	D
B. Routing packets between different networks	D. Managing MAC address assignments for local devices	

60) Which of the following network devices operates primarily at the Network Layer of the OSI model?

A. Switch	C. Hub	B
B. Router	D. Repeater	

61) What is the data unit used in the **Network Layer** of the OSI model?

A. Bit	C. Packet	C
B. Frame	D. Segment	

62) What key information is included in the header of a packet at the Network Layer ?

A. Source & Destination MAC addresses	C. Port numbers for applications	B
B. Source & Destination IP addresses	D. Error detection codes	

63) Which OSI layer is located **between** the Data Link Layer and the Transport Layer ?

A. Physical Layer	C. Session Layer	B
B. Network Layer	D. Presentation Layer	

64) What is the term for breaking large messages into smaller units at the **Transport Layer**?

A. Fragmentation	C. Multiplexing	B
B. Segmentation	D. Compression	

65) Which of the following OSI layers provide **error control mechanisms**?

A. Data Link Layer	D. A + B	D
B. Transport Layer	E. A + C	
C. Session Layer	F. B + C	



66) Which of the following are **functions of the Transport Layer** in the OSI model?

A. Flow control and error correction	D. A + B	E
B. Routing and logical addressing	E. A + C	
C. segmentation and Reassembly	F. B + C	

67) What is the **data unit** used in the **Transport Layer** of the OSI model?

A. Bit	C. Packet	D
B. Frame	D. Segment	

68) What is the **primary function** of port numbers in the **Transport Layer** ?

A. Identifying devices on a network	C. Determining the best route for packets across the network	B
B. Identifying specific applications or services on a device	D. Managing encryption and decryption of data	

69) How many bits are used to represent a **Port Number** in the Transport Layer?

A. 8 bits	C. 32 bits	B
B. 16 bits	D. 64 bits	

70) In which OSI layer are **port numbers** used as a form of addressing?

A. Data Link Layer	C. Transport Layer	C
B. Network Layer	D. Application Layer	

71) Which port number is used by the HTTP (Hypertext Transfer Protocol) for web traffic?

A. 21	C. 53	D
B. 25	D. 80	

72) What is the default port number for secure web traffic (HTTPS) ?

A. 22	C. 443	C
B. 53	D. 110	

73) Which port number is used by FTP (File Transfer Protocol) for file transfers?

A. 21	C. 110	A
B. 25	D. 143	

74) What is the primary responsibility of the Transport Layer?

a) Delivering data from one device to another	c) Delivering data from one network to another	B
b) Delivering data from one process to another	d) Delivering data from one frame to another	

75) What is used by the Transport Layer to identify the destination process?

a) IP address	c) Port number	C
b) MAC address	d) Sequence number	

76) What is the main **difference** between **Network Layer** delivery and **Transport Layer** delivery?

a) Network Layer delivers data from device to device, while Transport Layer delivers data from process to process.	c) Both layers deliver data from device to device.	A
b) Network Layer delivers data from process to process, while Transport Layer delivers data from device to device.	d) Both layers deliver data from process to process.	

77) What is the primary **function of the Session Layer**?

a) Encrypting data for security	c) Routing data between networks	B
b) Managing dialogs and synchronizing communication	d) Transmitting bits over a physical medium	

78) What does "**Dialog Control**" in the Session Layer **refer to**?

a) Managing the flow of data between applications	c) Routing data between networks	A
b) Encrypting data for secure communication	d) Transmitting bits over a physical medium	

79) Which layer is **responsible** for **synchronizing** the transmission of **bits** between devices?

a) Physical Layer	d) Transport Layer	A
b) Data Link Layer	e) a + b	
c) Session Layer	f) a + c	

80) Which layer is responsible for adding **synchronization** points to data streams to allow resuming **communication after a failure**?

a) Physical Layer	d) Transport Layer	C
b) Data Link Layer	e) a + b	
c) Session Layer	f) a + c	

81) Which of the following layers are responsible for **synchronization**?

a) Physical Layer	d) Transport Layer	F
b) Data Link Layer	e) a + b	
c) Session Layer	f) a + c	

82) What is the primary **function** of the **Presentation Layer**?

a) Routing data between networks	c) Transmitting bits over a physical medium	B
b) Translation, compression, and encryption.	d) Managing dialogs between applications	

ملاحظة:

يمكن ان تكون الإجابة أيضاً : **Formatting and translating data for applications**

83) Which of the following is a responsibility of the **Presentation Layer**?

a) Encrypting data for security	c) Detecting errors in transmitted frames	A
b) Assigning IP addresses to devices	d) Managing the flow of data between devices	

84) What does the **Presentation Layer** use to **reduce the size of data**?

a) Encryption	c) Translation	B
b) Compression	d) Synchronization	

85) What is the primary function of the **Application Layer**?

a) Routing data between networks	c) Transmitting bits over a physical medium	B
b) Providing an interface for users to access network services	d) Managing dialogs between applications	

86) Which layer is responsible for using **port numbers** to deliver data to the correct application?

a) Physical Layer	c) Transport Layer	C
b) Data Link Layer	d) Application Layer	

87) Which layer is responsible for handling **physical addresses** ?

a) Physical Layer	c) Network Layer	B
b) Data Link Layer	d) Transport Layer	

88) Which of the following is an example of a specific address used in the **Application Layer**?

a) MAC address	c) Port number	D
b) IP address	d) Email address	

89) Which of the following correctly matches the address type to its corresponding layer?

a) MAC address - Network Layer	c) Port number - Transport Layer	C
b) IP address - Data Link Layer	d) Email address - Physical Layer	

90) How many bits and bytes are in a MAC address?

a) 32 bits, 4 bytes	c) 64 bits, 8 bytes	B
b) 48 bits, 6 bytes	d) 128 bits, 16 bytes	

91) How many hexadecimal digits are used to represent a MAC address?

a) 8	c) 16	B
b) 12	d) 24	

92) What character is used to separate every two hexadecimal digits in a MAC address ?

a) . (dot)	c) - (hyphen)	B
b) : (colon)	d) / (slash)	

93) Which of the following is a valid MAC address?

a) 00/1A/2B/3C/4D/5E	c) 00:1A:2B:3C:4D:5E	C
b) 00.1A.2B.3C.4D.5G	d) 00-1A-2B-3C-4D-5E	

94) Which of the following is a **valid MAC address**?

a) 00:1A:2B:3C:4D:5H	c) 00:1A:2B:3C:4D:5E:6F:11	B
b) 00:1A:2B:3C:4D:5E	d) 00:1A:2B:3C:4D	

95) Which of the following statements is **true** about **MAC addresses**?

a) MAC addresses change at the device level but remain the same from hop to hop.	c) MAC addresses change both at the device level and from hop to hop.	B
b) MAC addresses remain the same at the device level but change from hop to hop.	d) MAC addresses remain the same both at the device level and from hop to hop.	

ملاحظة :

- عنوان **MAC** يبقى كما هو على مستوى الجهاز لأنه مرتبط بالجهاز نفسه.

- عنوان **MAC** يتغير من قفزة إلى قفزة لأن كل جهاز وسيط (مثل **Router** أو **Switch**) له عنوان **MAC** خاص به

96) Which of the following addresses **change from hop to hop**?

a) MAC address	b) IP address	A
b) IP address	d) Email address	

97) Which layer is responsible for handling addresses that remain the same from hop to hop?

a) Data Link Layer	c) Transport Layer	D
b) Network Layer	d) Both b and c	

ملاحظة:

Port number & IP address

98) Which of the following is a **valid** port number?

a) 65536	c) 1024.5	B
b) 8080	d) -1	

ملاحظة:

بالنسبة للخيار A هذا يعني أنه خارج النطاق المسموح.
أما بالنسبة لـ C لا يمكن أن تكون قيمة port number قيمة عشرية.
أما D لا يمكن أن يكون port number سالباً.

99) Which of the following is a valid representation of a port number?

a) 8080	c) 80.80	A
b) 80:80	d) 80-80	

100) Which of the following statements about network layers is **correct**?

a) The division of network layers is physical, not logical.	c) Companies can change inter-layer protocols but not intra-layer protocols.	B
b) The Application Layer is independent of how the message is transmitted.	d) The Physical Layer adds a header to the data.	

101) Which of the following is **true** about LAN cables?

a) LANs use only one type of cable.	c) LANs use only wireless connections.	B
b) LANs can use UTP, fiber optic, or coaxial cables.	d) LANs do not require cables.	



102) Which of the following statements about message transmission is correct?

a) The delay between layers is significant compared to transmission delay.	c) The Data Link Layer checks the message after each hop.	C
b) Messages are routed between intermediate nodes using MAC addresses.	d) The Physical Layer adds a header to the data.	

103) Which of the following is true about sockets?

a) A socket is a combination of MAC address and port number.	c) A socket is a type of cable used in networking.	B
b) A socket is a combination of IP address and port number.	d) A socket is a protocol used in the Application Layer.	

104) Which of the following is not true about LAN cables?

a) LANs can use UTP cables.	c) LANs can use coaxial cables.	D
b) LANs can use fiber optic cables.	d) LANs use only wireless connections.	

105) Which of the following statements about message transmission is not correct?

a) The delay between layers is small compared to transmission delay.	c) The Data Link Layer checks the message after each hop.	D
b) Messages are routed between intermediate nodes using IP addresses.	d) The Physical Layer adds a header to the data.	

تأتي الأشياء على قدر سعيك لا على قدر رغبتك



106) Which of the following is not true about sockets?

a) A socket is a combination of IP address and port number.	c) A socket is a protocol used in the Application Layer.	C
b) A socket is used to identify a specific connection between applications.	d) A socket is essential for establishing communication between devices.	

107) What is a protocol in networking?

a) A physical device used to connect computers	c) A type of cable used for data transmission	B
b) A set of rules that govern data communication	d) A software application for managing networks	

108) What is the difference between inter-layer protocols and intra-layer protocols?

a) Inter-layer protocols govern communication between layers, while intra-layer protocols govern communication within a single layer.	c) Inter-layer protocols are developed by companies, while intra-layer protocols are standardized.	A
b) Inter-layer protocols are used only in wireless networks, while intra-layer protocols are used in wired networks.	d) Inter-layer protocols are used for error detection, while intra-layer protocols are used for data encryption.	

ملاحظة :

(Inter-Layer Protocols): هي قواعد أثناء الانتقال بين الطبقات

(Intra-Layer Protocols): هي قواعد بداخل الطبقة

109) The _____ model shows how the network functions of a computer ought to be organized.

A. CCITT	C. IOS	B
B. OSI	D. ANSI	

110) As the data packet moves from the upper to the lower layers, headers are _____.

A. Added

C. Rearranged

B. Removed

D. Modified

A

111) When a host on network A sends a message to a host on network B, which address does the **router** look at?

A. Port

C. physical

B. Logical

D. none of the above

B

112) To deliver a message to the correct application program running on a host, the _____ address must be consulted.

A. Port

C. physical

B. IP

D. none of the above

A

113) The _____ layer adds a header to the packet coming from the upper layer that includes the logical addresses of the sender and receiver.

A. physical

C. network

B. data link

D. none of the above

C

114) The physical, data link, and network layers are the _____ support layers.

A. User

C. both A and B

B. Network

D. neither A nor B

B

115) The session, presentation, and application layers are the ____ support layers.

A. User	C. both A and B	A
B. Network	D. neither A nor B	

116) The ____ layer links the network support layers and the user support layers

A. Transport	C. Data Link	A
B. Network	D. Session	

117) The ____ layer is responsible for delivering data units from one station to the next without errors

A. Transport	C. Data Link	C
B. Network	D. Session	

118) The TCP/IP ____ layer is equivalent to the combined session, presentation, and application layers of the OSI model.

A. Application	C. Data Link	A
B. Network	D. physical	

119) The ____ address uniquely defines a host on the Internet

A. physical	C. Port	B
B. IP	D. specific	

120) The _____ address identifies a process on a host.

A. physical

C. Port

B. IP

D. specific

C

121) The _____ layer is the layer closest to the transmission medium.

A. Physical

C. Network

B. Data Link

D. Transport

A

122) Mail services are available to network users through the _____ layer.

A. Data Link

C. Transport

B. Physical

D. Application

D

123) Which of the following is an application layer service ?

A. Remote log-in

C. Mail service

B. File transfer and access

D. All the above

D

124) When a data packet moves from the lower layers to the upper layers, the headers are _____.

A. Added

C. Rearranged

B. Removed

D. Modified

B



Wired LANs: Ethernet

1) What was the primary goal of IEEE Project 802?

A. To develop a new internet protocol	C. To improve wireless security protocols	B
B. To standardize communication between different network devices	D. To replace TCP/IP with a better alternative	

2) Which OSI layer is covered by IEEE Project 802

A. Physical Layer	C. Data Link Layer	E
B. Network Layer	D. A&B	
E. A&C	F.	

3) The Data Link Layer in IEEE 802 is divided into which two sublayers?

A. MAC and IP	C. TCP and UDP	B
B. LLC and MAC	D. Frame Relay and ATM	

4) What is the primary responsibility of the MAC sublayer in IEEE 802 networks?

A. Controlling access to the transmission medium and managing frame placement	C. Assigning logical addresses to devices on the network	A
B. Converting data into electrical or optical signals for transmission	D. Establishing and terminating network connections between hosts	



5) Which IEEE standard defines Ethernet networking?

A. IEEE 802.3	C. IEEE 802.15	A
B. IEEE 802.11	D. IEEE 802.1X	

6) The IEEE standard for wireless networking (Wi-Fi) is:

A. IEEE 802.1	C. IEEE 802.11	C
B. IEEE 802.3	D. IEEE 802.15	

7) Which of the following fields are included in an HDLC frame?

A. Address	C. Data (Payload)	E
B. Control	D. FCS (Frame Check Sequence)	
E. All of the above	F. 	

8) What is the purpose of the Frame Check Sequence (FCS) field in an HDLC frame?

A. It specifies the address of the sender	C. It indicates the end of the frame	B
B. It provides error detection using CRC	D. It determines the frame length	

9) What is the correct term for the protocols that define how Ethernet operates between the Transmission Medium and the Physical Layer?

A. Ethernet MAC Standards	C. Ethernet LLC Standards	B
B. Ethernet Physical layers Standards	D. All of the above	

10) What is the main role of Ethernet PHY Standards?

A. Defining how data is framed and addressed	C. Defining how signals are transmitted over the physical medium	C
B. Managing logical link control functions	D. Assigning IP addresses to network devices	

11) Which fields does the LLC (Logical Link Control) sublayer add to the data before passing it to the MAC sublayer?

A. Source MAC Address, Destination MAC Address, and Frame Check Sequence (FCS)	C. IP Header, TCP Header, and Payload	B
B. DSAP (Destination Service Access Point), SSAP (Source Service Access Point), and Control	D. Type/Length Field, Preamble, and Start Frame Delimiter	

12) What happens to the LLC PDU (Protocol Data Unit) after it is created by the LLC sublayer?

A. It is directly transmitted over the physical medium	C. It is converted into an IP Packet and sent to the Network Layer	B
B. It is encapsulated inside a MAC Frame before transmission	D. It is used to manage TCP connections	

13) What is the direct output of the LLC sublayer before it is passed to the MAC sublayer?

A. LLC PDU	C. Ethernet Frame	A
B. MAC Payload	D. TCP Segment	

14) Which of the following are the **main components of an Ethernet MAC frame** according to IEEE 802.3?

A. Ethernet Header, Frame Body, Trailer, TCP Checksum	C. Source IP Address, Destination IP Address, MAC Header, Frame Check Sequence	D
B. Start Frame Delimiter (SFD), IP Header, TCP Header, Payload, CRC	D. Preamble, SFD, Destination MAC Address, Source MAC Address, Length/Type, Data, FCS	

15) How many **generations of Ethernet evolution** are officially recognized based on speed and technological advancements?

A. Three generations	C. Five generations	B
B. Four generations	D. Six generations	

ملاحظة:

Ethernet evolution : Standard Ethernet / Fast Ethernet / Gigabit Ethernet / Ten-Gigabit Ethernet

16) Which part of an Ethernet frame is responsible for **synchronizing the communication** between sender and receiver?

A. CRC	C. Source and Destination MAC Addresses	D
B. SFD(Start Frame delimiter)	D. Preamble	

17) What is the minimum payload size allowed in an Ethernet frame?

A. 128 bits	C. 46 Bytes	C
B. 32 Bytes	D. 600 bits	

18) What is the **minimum valid Ethernet frame size** according to IEEE 802.3?

A. 32 Bytes

C. 1500 bits

B. 64 Bytes

D. 1500 Bytes

B

19) What is the **maximum payload size** allowed in an Ethernet frame?

A. 16 Bytes

C. 1500 bits

A. 64 Bytes

D. 1500 Bytes

D

20) What is the **maximum allowable Ethernet frame size** according to IEEE 802.3?

A. 1516 Bytes

C. 1518 Bytes

B. 1500 Bytes

D. 1516 bits

C

21) What is the total length of an Ethernet frame excluding the payload, but including all required headers and trailers?

A. 12 Bytes

C. 22 Bytes

B. 18 Bytes

D. 26 Bytes

B

22) Which of the following must be true for an Ethernet address to be considered Unicast?

A. It must contain only numbers (0-9)

C. The least significant bit (LSB) of the first byte must be 0

B. The least significant bit (LSB) of the first byte must be 1

D. It must be exactly 64 bits long

C

23) In an Ethernet MAC address, how do we determine if an address is Unicast or Multicast?

- | | |
|--|--|
| A. By checking if the first byte is even or odd | C. By checking if all bytes are set to 1s |
| B. By checking the least significant bit (LSB) of the first byte | D. By examining the Frame Check Sequence (FCS) |

B

24) What is the key difference between Multicast and Broadcast in Ethernet?

- | | |
|---|---|
| A. Multicast sends frames to all devices in the network, while Broadcast sends frames to a selected group | C. Broadcast sends frames to all devices, while Multicast sends frames to a specific group of devices |
| B. Multicast uses the address FF:FF:FF:FF:FF:FF, while Broadcast does not | D. Multicast addresses always have their first byte set to 00 |

C

ملاحظة:

لتحديد نوع العنوان ننظر إلى الرقم الثاني من اليسار (لـ العنوان الممثل بـ النظام السداسي عشر):

- إذا كان زوجي يكون العنوان أحادي الإرسال
- وإذا كان فردي يكون العنوان متعدد الإرسال.
- وإذا كانت الأرقام كلها F يكون العنوان broadcast

25) What type of MAC address is 4A:30:10:21:10:1A?

- | | |
|--------------|-------------------------|
| A. Unicast | C. Broadcast |
| B. Multicast | D. Cannot be determined |

A

توضيح:

4A:30:10:21:10:1A > A :1010



26) Given the MAC address 47:20:1B:2E:08:EE, what type of address is it?

- | | |
|--------------|--------------|
| A. Unicast | C. Broadcast |
| B. Multicast | D. Reserved |

B

توضيح:

47:20:1B:2E:08:EE > 7:0111

27) Which of the following MAC addresses is a valid Unicast address?

- | | |
|----------------------|----------------------|
| A. 3B:45:67:89:AB:CD | C. 57:12:34:56:78:9A |
| B. 2A:34:56:78:9A:BC | D. FF:FF:FF:FF:FF:FF |

B

توضيح:

لأن A = 1010 و LSB = 0

28) Which MAC address is a Multicast address?

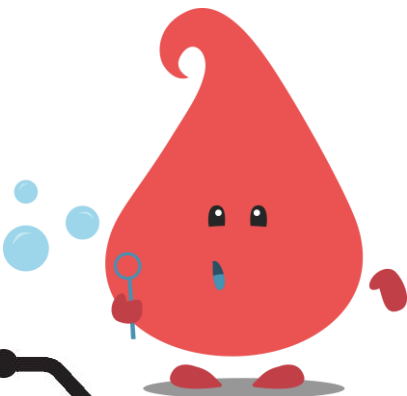
- | | |
|----------------------|----------------------|
| A. 48:1A:2B:3C:4D:5E | C. AA:BB:CC:DD:EE:FF |
| B. 59:67:89:AB:CD:EF | D. 20:15:9A:BC:DE:F0 |

B

29) Which of the following MAC addresses is not a Unicast address?

- | | |
|----------------------|----------------------|
| A. 24:15:9A:BC:DE:F0 | C. 35:45:67:89:AB:CD |
| B. 50:67:89:AB:CD:EF | D. C0:12:34:56:78:90 |

C



30) Given the MAC address 47:20:1B:2E:08:EE, how will it be transmitted on the network based on Ethernet LSB-first transmission?

- A. 01000111 00100000 00011011 00101110 00001000 11101110
- B. 11100010 00000100 11011000 01110100 00010000 01110111
- C. 11100010 00100000 00011011 00101110 00001000 11101110
- D. 01000111 00000100 11011000 01110100 00010000 01110111

B

توضيح:

يتم إرسال البايتات بترتيبها الطبيعي **من اليسار إلى اليمين**.

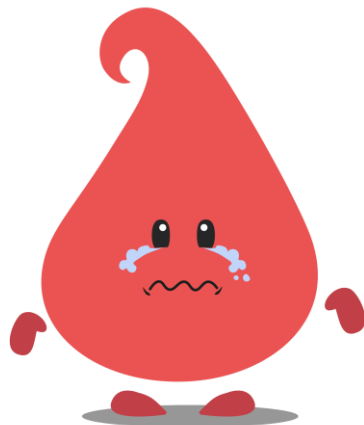
47 : 20 : 1B : 2E : 08 : EE
1 2 3 4 5 6
→

لكن **داخل** كل بايت، يتم إرسال البتات **من اليمين إلى اليسار**

31) The MAC address A5:3C:7D:12:90:EF is transmitted on an Ethernet network. What will be the correct bit sequence on the line?

- A. 10100101 00111100 10111110 01001000 00001001 11110111
- B. 10100101 00111100 01111101 00010010 10010000 11101111
- C. 10100101 00111100 10111110 01001000 00001001 01110111
- D. 01011010 00111100 10111110 01001000 00001001 11110111

A



**بقي القليل حتى تفرج المذن ،
أرأيت ممتحننا يمضي العمر يمتحن؟**

32) When an Ethernet MAC address is transmitted on the network, how are its bits and bytes sent on the wire?

- | | |
|---|--|
| A. Bytes are sent right-to-left, and bits within each byte are sent left-to-right | C. Both bytes and bits within each byte are sent right-to-left |
| B. Bytes are sent left-to-right, and bits within each byte are sent right-to-left | D. Both bytes and bits within each byte are sent left-to-right |

B

33) Which of the following statements is true regarding Switched Ethernet compared to Bridged Ethernet?

- | | |
|---|--|
| A. Switched Ethernet creates more collisions than Bridged Ethernet. | C. Switched Ethernet cannot operate in full-duplex mode. |
| B. Switched Ethernet eliminates collisions by creating dedicated virtual links between sender and receiver. | D. Bridged Ethernet is more efficient than Switched Ethernet in high-speed networks. |

B

34) Why were traditional telephone cables unable to handle Fast Ethernet?

- | | |
|--------------------------------|---|
| a) They were too expensive | c) They were not compatible with Ethernet |
| b) They were affected by noise | d) They were too thick |

B

35) What happens to the noise resistance of twisted pair cables as the number of twists increases?

- | | |
|------------------------|-----------------------------|
| a) It decreases | C) It Increase |
| b) It remains the same | d) It becomes unpredictable |

C

توضيح:

في كبلات الـ **twisted pair** كل ما زادت عدد الجدلات بالكبل يقل تأثيره بالضجيج لكن مقاومة الضجيج تزداد.

36) Which type of twisted pair cable was used when speeds reached 100 MHz?

a) Twisted Pair 3

c) Twisted Pair 5

b) Twisted Pair 4

d) Twisted Pair 6

C

37) Which IEEE standard defines Fast Ethernet?

a) IEEE 802.3

c) IEEE 802.3z

b) IEEE 802.3u

d) IEEE 802.3ae

B

38) What is the speed of Fast Ethernet?

a) 10 Mbps

c) 1 Gbps

b) 100 Mbps

d) 10 Gbps

B

39) Which Ethernet standard supports 10 Gbps speeds?

a) IEEE 802.3

c) IEEE 802.3z

b) IEEE 802.3u

d) IEEE 802.3ae

D

40) Which IEEE standard defines Gigabit Ethernet over Twisted Pair cables?

a) IEEE 802.3z

c) IEEE 802.3u

b) IEEE 802.3ab

d) IEEE 802.3ae

B

41) Gigabit Ethernet has a data rate of _____ Mbps

- | | |
|--------|----------|
| A. 10 | C. 1000 |
| B. 100 | D. 10000 |

C

42) _____ defines a protocol data unit (PDU) that is somewhat similar to that of HDLC

- | | |
|--------|---------|
| A. MAC | C. LLU |
| B. LLC | D. None |

B

43) In the Ethernet, the _____ field is actually added at the physical layer and is not (formally) part of the frame.

- | | |
|-------------|------------|
| A. CRC | C. address |
| B. Preamble | D. none |

B

44) Standard Ethernet uses _____ encoding

- | | |
|--------|----------------------------|
| A. NRZ | C. Manchester |
| B. AMI | D. differential Manchester |

C

45) Which of the following could not be an Ethernet unicast destination?

- | | |
|----------------------|----------------------|
| A. 43:7B:6C:DE:10:00 | C. 46:56:21:1A:DE:F4 |
| B. 44:AA:C1:23:45:32 | D. 48:32:21:21:4D:34 |

A

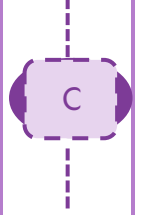
46) Which of the following could not be an Ethernet multicast destination?

A. B7:7B:6C:DE:10:00

C. 7C:56:21:1A:DE:F4

B. 7B:AA:C1:23:45:32

D. 83:32:21:21:4D:34



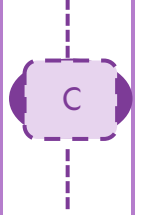
47) Each station on an Ethernet network has a unique _____ address imprinted on its network interface card (NIC).

A. 5-byte

C. 48-bit

B. 32-bit

D. none of the above



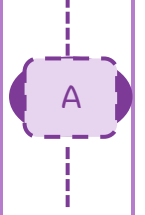
48) What is the throughput efficiency of the ALOHA protocol?

a) 18%

c) 50%

b) 36%

d) 75%



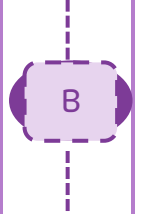
49) What is the throughput efficiency of the Slotted ALOHA protocol?

a) 18%

c) 50%

b) 36%

d) 75%



إذا كانت نيتك لا أبرح حتى أبلغ..

فلا بد أن تصبر على ما لم تُحط به خيرا



Switching

1) Which of the following statements about network switches is TRUE?

A) Switches operate in half-duplex mode by default.	C) Switches use circuit-switching to establish a dedicated path for each communication.	B
B) Switches always work in full-duplex mode, allowing simultaneous sending and receiving of data.	D) Switches require manual configuration for every new device connected to them.	

2) Which of the following is NOT a characteristic of a circuit-switched network?

A) A dedicated path is established before communication starts.	C) Packets are sent independently without reserving a path.	C
B) Resources are reserved for the entire duration of communication.	D) It is commonly used in traditional telephone networks.	

3) How many main types of switched networks exist according to the taxonomy of switched networks?

A) 2	C) 4	B
B) 3	D) 5	

4) Which of the following is NOT a correct type of switched networks?

A) Circuit-Switched Networks	C) Hybrid-Switched Networks	C
B) Packet-Switched Networks	D) Virtual-Circuit Networks	



5) Which of the following statements about circuit-switched networks is TRUE?

A) Data packets are sent independently using different paths.	C) Circuit switching is more efficient than packet switching in utilizing network resources.	B
B) A dedicated path is reserved for the entire communication session.	D) Resources are allocated dynamically based on demand.	

6) Which of the following is NOT a characteristic of circuit-switched networks?

A) The network establishes a dedicated connection before data transfer begins.	C) Data packets choose the most efficient path dynamically.	C
B) Each link is divided into multiple channels using FDM or TDM.	D) Circuit-switched networks are commonly used in traditional telephone systems.	

7) Which of the following is a unique characteristic of circuit-switched networks compared to other types of switched networks?

A) They establish a dedicated path for the entire communication session.	C) They use dynamic routing to find the best path for each packet.	A
B) They divide data into packets that travel independently.	D) They allow packets from different sources to share the same path dynamically.	

8) In circuit-switched networks, what type of connection is established between two communicating stations?

A) A logical path that can change dynamically	C) A best-effort connection without resource reservation	B
B) A dedicated physical link reserved for the entire session	D) A shared path used by multiple devices simultaneously	

9) In circuit-switched networks, how are links between switches utilized?

A) Each link is dynamically shared between multiple connections in real time.

B) Each connection gets a dedicated channel on every physical link it uses.

C) Links are used only when packets need to be transmitted, freeing resources dynamically.

D) Each link can change its assigned connection path based on traffic conditions.

B

10) What happens to network resources in a circuit-switched network once a connection is established?

A) Resources are dynamically allocated based on demand.

B) Resources are shared among multiple users during the connection.

C) Resources are only used when data is actively being transmitted.

D) Resources are reserved for the entire duration of the communication.

D

11) Which of the following is a key difference between circuit-switched networks and packet-switched networks?

A) Circuit-switched networks require resource reservation, while packet-switched networks allocate resources on demand.

B) Circuit-switched networks use packet-based data transmission, while packet-switched networks use continuous data streams.

C) Circuit-switched networks are more efficient in bandwidth utilization than packet-switched networks.

D) Circuit-switched networks allow dynamic routing of data, while packet-switched networks use fixed paths.

A

12) Which of the following are the three main phases of a circuit-switched network connection?

A) Routing Phase, Transmission Phase, Termination Phase

B) Setup Phase, Data Transfer Phase, Teardown Phase

C) Packet Forwarding Phase, Path Selection Phase, Session Termination Phase

D) Addressing Phase, Connection Establishment Phase, Acknowledgment Phase

B

13) What happens during the setup phase in a circuit-switched network?

- | | |
|--|--|
| A) The connection is terminated, and resources are released. | C) Data packets are forwarded dynamically to the destination |
| B) The network determines the shortest available path for each packet. | D) A path is established between the sender and receiver, and resources are reserved |

D

14) What is the purpose of the teardown phase in a circuit-switched network?

- | | |
|--|---|
| A) To maintain the dedicated path for future use. | C) To establish a new connection with another receiver. |
| B) To release the reserved network resources after communication ends. | D) To improve the quality of service by reallocating bandwidth dynamically. |

B

15) What is the typical data transmission rate of a T-1 connection?

- | | |
|---------------|-------------|
| A) 64 Kbps | C) 10 Mbps |
| B) 1.544 Mbps | D) 100 Mbps |

B

16) In a circuit-switched network, which of the following sequences correctly describes the phases of communication?

- | | |
|--|--|
| a) Transmission → Connect → Disconnect | c) Disconnect → Connect → Transmission |
| b) Connect → Transmission → Disconnect | d) Transmission → Disconnect → Connect |

B

17) Which layer of the OSI model is responsible for circuit switching in the traditional telephone network?

- | | |
|--------------------|--------------------|
| a) Data Link Layer | c) Physical Layer |
| b) Network Layer | d) Transport Layer |

C

18) Why is circuit switching suitable for traditional telephone networks?

a) It allows dynamic sharing of resources.	c) It is more efficient for bursty data traffic.	B
b) It provides a continuous and reliable connection.	d) It reduces the need for physical infrastructure.	

19) In the traditional telephone network, which switching approach is used at the physical layer?

a) Packet Switching	c) Circuit Switching	C
b) Message Switching	d) Virtual Circuit Switching	

20) In a datagram network, how are packets routed to their destination?

a) A dedicated path is established before sending packets.	c) Packets are sent in a predefined sequence.	B
b) Each packet is routed independently based on network conditions.	d) All packets follow the same path to the destination.	

21) Which of the following is a characteristic of datagram networks?

a) Guaranteed delivery of packets in order.	c) Packets may arrive out of order at the destination.	C
b) A dedicated path is established for each communication session.	d) Resources are reserved for the entire duration of communication.	



22) What type of data is best suited for datagram networks?

- | | |
|---|--|
| a) Continuous data streams, like voice calls. | c) High-priority real-time data. |
| b) Bursty data, like web browsing or email. | d) Data requiring guaranteed delivery. |

توضيح :

Switching in the Internet is done by using the datagram approach

23) Which of the following is NOT a feature of datagram networks?

- | | |
|---|---|
| a) Packets may take different paths to the destination. | c) Packets are routed independently. |
| b) A dedicated path is established before communication starts. | d) Resources are shared among multiple users. |

24) In a datagram network, which of the following statements about packets arriving at the destination is INCORRECT?

- | | |
|---|---|
| a) Packets may arrive out of order. | c) Packets always follow the same path. |
| b) Packets may be lost during transmission. | d) Packets may experience different delays. |

25) Which two factors primarily determine how packets are segmented in a network?

- | | |
|--|---|
| a) Network speed and number of users. | c) Application type and network congestion. |
| b) Network MTU and governing protocol. | d) Packet size and destination address. |

26) In packet segmentation, packets can be divided into:

- | | |
|-------------------------|--------------------------------------|
| a) Fixed sizes only. | c) Either fixed or variable sizes. |
| b) Variable sizes only. | d) Neither fixed nor variable sizes. |

27) In a packet-switched network, how are resources allocated?

- | | |
|--|---|
| a) Resources are reserved before communication starts. | c) Resources are shared equally among all users at all times. |
| b) Resources are allocated on demand during communication. | d) Resources are allocated based on a fixed schedule |

B

28) In a packet-switched network, which of the following statements about packets is TRUE?

- | | |
|--|--|
| a) Packets always arrive in the same order they were sent. | c) Packets may arrive out of order, and their delivery is not guaranteed. |
| b) Packets are guaranteed to arrive without any loss. | d) Packets are always reassembled at the destination in their original size. |

C

29) In a datagram network, what does a switch use to determine the next hop for a packet?

- | | |
|--|---|
| a) Source address in the packet header. | c) Size of the packet. |
| b) Destination address in the packet header. | d) Time-to-live (TTL) field in the packet header. |

B

30) Which of the following statements about datagram networks is FALSE?

- | | |
|--|--|
| a) Each packet is routed independently. | c) Switches use routing tables to forward packets. |
| b) The destination address may change during the packet's journey. | d) Packets may arrive out of order at the destination. |

B

33) Which approach to packet switching is used in the Internet at the network layer?

- | | |
|------------------------------|-----------------------|
| a) Circuit Switching | c) Datagram Switching |
| b) Virtual Circuit Switching | d) Message Switching |

C

34) At which layer of the OSI model is datagram switching primarily implemented in the Internet?

a) Physical Layer

c) Network Layer

b) Data Link Layer

d) Transport Layer

C

35) Which of the following is NOT a phase in Virtual-Circuit Switching?

A) Setup phase

C) Resource reservation phase

B) Data transfer phase

D) Teardown phase

C

36) What is the role of the Virtual Circuit Identifier (VCI)?

A) Identifies the final destination of a packet

C) Ensures that packets arrive at the same time

B) Helps routers determine the next hop in a virtual circuit network

D) Dynamically assigns bandwidth to each connection

B

37) Match the following network switching types is true

A) The Circuit-Switched Networks is a hybrid between the Datagram Networks and Virtual-Circuit Networks

C) The Virtual-Circuit Networks is a hybrid between the Circuit-Switched Networks and Datagram Networks

B) The Datagram Networks is a hybrid between the Circuit-Switched Networks and Virtual-Circuit Networks.

D) none of the above

C

38) What is a key characteristic of Virtual-Circuit Switching?

A) Packets from the same source and destination may take different paths

C) Packets follow the same path but may experience different delays

B) Packets always arrive at the destination with the same delay

D) Resources are permanently reserved for each connection

C

39) Why might packets in a Virtual-Circuit Network experience different delays?

- | | | |
|---|---|---|
| A) Because they take different routes | C) Because they contain different amounts of data | B |
| B) Due to on-demand resource allocation | D) Due to fixed bandwidth reservation | |

40) How does Virtual-Circuit Switching differ from Circuit-Switched Networks in terms of delay?

- | | | |
|--|--|---|
| A) Virtual-Circuit Switching has a constant delay like Circuit-Switched Networks | C) Circuit-Switched Networks have reserved resources, leading to consistent delays | C |
| B) Virtual-Circuit Switching allows packets to take different paths | D) Virtual-Circuit Switching does not require a setup phase | |

42) What is a key difference between Virtual-Circuit Networks and Datagram Networks?

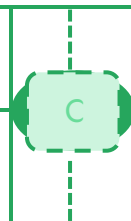
- | | | |
|---|---|---|
| A) Virtual-Circuit Networks require a setup phase before communication | C) Datagram Networks use a dedicated path for the entire connection | A |
| B) Virtual-Circuit Networks allow packets to take different paths dynamically | D) Virtual-Circuit Networks do not require routing tables | |

43) Which of the following best describes the key difference in how destination information is handled in Circuit-Switched Networks versus Virtual-Circuit Networks?

- | | | |
|---|---|---|
| A) In Circuit-Switched Networks, the destination address remains constant, whereas in Virtual-Circuit Networks, the VCI changes at each switch. | C) In both Circuit-Switched and Virtual-Circuit Networks, the destination address remains unchanged throughout the communication. | A |
| B) In Circuit-Switched Networks, the destination address changes at each hop, whereas in Virtual-Circuit Networks, the VCI remains constant throughout the network. | D) In Virtual-Circuit Networks, the VCI remains unchanged throughout the network, just like the destination address in Circuit-Switched Networks. | |

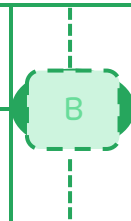
44) What information is stored in a switch table of a Virtual-Circuit Network?

- | | |
|--|---|
| A) The destination IP address of each packet | C) The mapping of incoming VCI to outgoing VCI and port |
| B) The MAC address of each device in the network | D) The list of all active TCP connections |



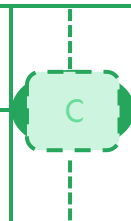
45) Which of the following is a key benefit of Virtual-Circuit Networks in data transfer?

- | | |
|---|---|
| A) Packets always arrive at the destination with the same delay | C) No setup phase is required for data transfer |
| B) Packets follow a predefined path and arrive in order | D) Resources are permanently reserved for each connection |



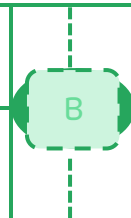
46) What happens during the setup phase in Virtual-Circuit Networks?

- | | |
|---|--|
| A) The network assigns a permanent dedicated path for communication | C) The network establishes a path and assigns VCI values for routing |
| B) The source device starts sending packets immediately | D) Packets are routed dynamically using destination IP addresses |



47) Why is the setup acknowledgment phase important in Virtual-Circuit Networks?

- | | |
|--|--|
| A) It reserves bandwidth permanently for the connection | C) It dynamically changes the path based on network congestion |
| B) It confirms that the path is established before data transfer | D) It allows packets to take different paths in the network |



48) Which of the following statements about packet ordering is TRUE?

- | | |
|--|--|
| A) Packets in Virtual-Circuit Networks always follow the same path, but they may arrive out of order due to dynamic resource allocation. | C) Packets in Virtual-Circuit Networks always arrive in order because they follow a predetermined path. |
| B) Datagram Networks ensure that packets arrive in order by reassembling them at each switch before forwarding. | D) Datagram Networks use a fixed path for each connection, ensuring that packets always arrive in order. |



49) Which of the following statements about packet delivery guarantees is FALSE?

A) Application Layer	C) Data Link Layer	C
B) Network Layer	D) Transport Layer	

50) At which OSI layer is Virtual-Circuit Switching typically implemented?

A) Application Layer	C) Data Link Layer	C
B) Network Layer	D) Transport Layer	

51) In which type of network is Virtual-Circuit Switching primarily used?

A) Local Area Networks (LANs)	C) Personal Area Networks (PANs)	B
B) Wide Area Networks (WANs)	D) Metropolitan Area Networks (MANs)	

52) Which type of network requires a setup delay before data transmission?

A) Virtual-Circuit Switching	C) Datagram Networks	D
B) Circuit-Switched Networks	D) Both A & B	

53) Which type of network suffers from unpredictable transmission delay?

A) Circuit-Switched Networks	C) Virtual-Circuit Networks	B
B) Datagram Networks	D) None of the above	

54) Which type of network provides consistent and predictable transmission delay?

A) Datagram Networks	C) Virtual-Circuit Networks	B
B) Circuit-Switched Networks	D) All of the above	

55) Why is delay in Virtual-Circuit Networks more stable than in Datagram Networks?

- | | |
|--|--|
| A) Because all packets follow the same predefined path. | C) Because packets are always transmitted at the same speed. |
| B) Because resources are permanently reserved for each connection. | D) Because there is no setup phase. |

A

56) What is the purpose of a crosspoint in a crossbar switch?

- | | |
|--------------------------------------|---------------------------------------|
| a) To store packets temporarily. | c) To prioritize certain connections. |
| b) To connect an input to an output. | d) To manage network congestion. |

B

57) What happens if two inputs request a connection to the same output in a crossbar switch?

- | | |
|---|---|
| a) Both connections are established simultaneously. | c) The switch uses arbitration to prioritize one connection. |
| b) The switch drops both connection requests. | d) The switch creates a virtual circuit for both connections. |

C

58) What is the primary advantage of a multistage switch compared to a crossbar switch?

- | | |
|--|-------------------------------------|
| a) It provides faster data transmission. | c) It guarantees no packet loss. |
| b) It reduces the number of crosspoints. | d) It simplifies network management |

B

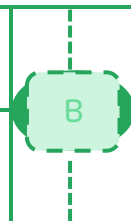
59) In a multistage switch, what is the purpose of the intermediate stages?

- | | |
|---|---|
| a) To connect inputs directly to outputs. | c) To store packets temporarily. |
| b) To prioritize certain connections | d) To route data between the input and output stages. |

D

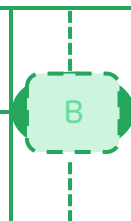
60) Which of the following statements about multistage switches is FALSE?

- | | |
|--|---|
| a) They use multiple stages to connect inputs to outputs | c) They are more scalable for large networks. |
| b) They require more crosspoints than crossbar switches. | d) They can handle multiple simultaneous connections. |



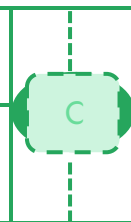
61) In a three-stage switch, what does the variable k represent?

- | | |
|--|--|
| a) The total number of inputs or outputs. | c) The number of inputs or outputs per switch in the first or third stage. |
| b) The number of switches in the intermediate stage. | d) The total number of crosspoints. |



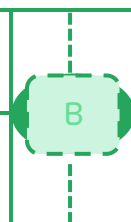
62) If $N=16, n=4, k=4$ what is the total number of crosspoints in a three-stage switch?

- | | |
|--------|--------|
| a) 128 | c) 192 |
| b) 256 | d) 512 |



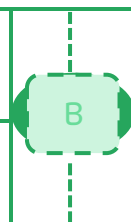
63) What is the main advantage of reducing the number of crosspoints in a switch?

- | | |
|--|--------------------------------------|
| a) It increases the speed of data transmission. | c) It guarantees no packet loss. |
| b) It reduces the cost and complexity of the switch. | d) It simplifies network management. |



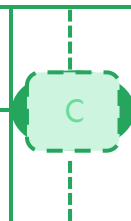
64) If $N=32, n=8, k=4$, what is the total number of crosspoints in a three-stage switch?

- | | |
|--------|--------|
| a) 512 | c) 640 |
| b) 320 | d) 704 |



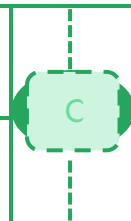
65) What is the primary advantage of using a Clos network in switch design?

- | | |
|--|---|
| a) It always guarantees the minimum number of crosspoints. | c) It optimizes crosspoint usage while allowing non-blocking communication. |
| b) It eliminates the need for a middle stage. | d) It only works for small-scale networks. |



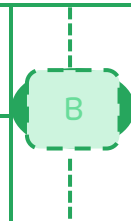
66) According to the Clos criterion, what is the relationship between k and n ?

- | | |
|-------------|-----------------|
| a) $k > n$ | c) $k > 2n - 1$ |
| b) $k > 2n$ | d) $k > n^2$ |



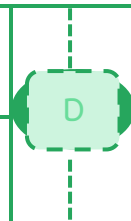
67) If $n=5$, what is the value of k according to the Clos criterion?

- | | |
|------|-------|
| a) 5 | c) 10 |
| b) 9 | d) 25 |



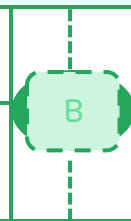
68) If $N=32$ and $n=8$, what is the value of k according to the Clos criterion?

- | | |
|-------|-------|
| a) 8 | c) 16 |
| b) 14 | d) 15 |

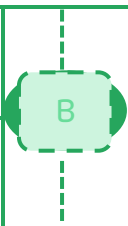


69) Using the values $N=16$, $n=4$ and $k=7$, what is the total number of crosspoints in a three-stage switch?

- | | |
|--------|--------|
| a) 224 | c) 448 |
| b) 336 | d) 560 |



70) According to the Clos criterion, if $N=32$, what is the value of n ?

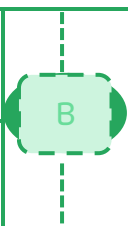
a) 2	c) 6	
b) 4	d) 8	

71) Which type of switch is generally the fastest in terms of data transmission speed?

a) Circuit Switch	c) Message Switch	
b) Packet Switch	d) Virtual Circuit Switch	

مسألة:

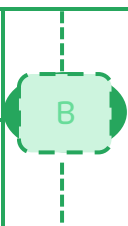
72) what is the value of n used to design the three-stage, 128×128 switch?

a) 6	c) 10	
b) 8	d) 12	

توضيح:

$$n = \left(\frac{N}{2}\right)^{1/2} = \sqrt{\frac{128}{2}} = 8$$

73) what is the value of k calculated using the Clos criterion?

a) 13	c) 17	
b) 15	d) 19	

توضيح:

$$K = 2n - 1 \quad K = 16 - 1 = 15$$

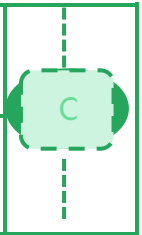
74) how many crossbars are there in the first stage of the three-stage switch?

a) 14

c) 16

b) 15

d) 17



توضيح:

$$\begin{aligned} \text{Crossbars in the first stage} &= N/n \\ &= 128/8 = 16 \end{aligned}$$

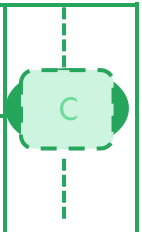
75) how many crossbars are there in the second stage of the three-stage switch?

a) 13

c) 15

b) 14

d) 16



توضيح:

$$\text{Crossbars in the second stage} = k = 15$$

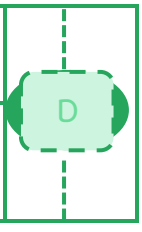
76) what is the total number of crosspoints in the three-stage switch?

a) 4500

c) 5000

b) 4800

d) 7680



توضيح:

$$\begin{aligned} \text{Crosspoints in the first stage} &= \text{Crossbars in the third stage} \\ &= n * k * \text{crossbars in stage} \end{aligned}$$

$$\text{Crosspoints in the first stage} = \text{Crossbars in the third stage} = 8 * 15 * 16$$

$$\text{Crosspoints in the first stage} = \text{Crossbars in the third stage} = 1920$$

$$\text{Crosspoints in the second stage} = N/n * N/n * \text{crossbars in stage}$$

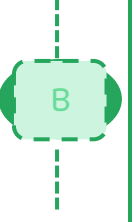
$$\text{Crosspoints in the second stage} = 16 * 16 * 15$$

$$\text{Crosspoints in the second stage} = 3840$$

total number of crosspoints in the three – stage switch

$$= 1920 + 960 + 1920 = 4800$$

77) which stage has the highest number of crosspoints?

a) First stage	c) Third stage	
b) Second stage	d) Both first and third stages	



ولقد يُنَالُكَ فِي الطَّرِيقِ مَشَقَّةٌ لَكِنَّ أَسْقَامَ الْجَهَالَةِ أَوْجَعُ

Logical Addressing

1- An IPv4 address consists of _____ bits.

A-4	C-32	C
B-8	D-64	

2- In IPv4, class _____ has the greatest number of addresses in each block

A-A	C-C	A
B-B	D-D	

يتم تحديد الكلاس الذي يندرج تحته ip address معين بناء على اول بايت

class	First byte	Default subnet mask
A	0----->127	0 /8
B	128----->191	10 /16
C	192----->223	110 /24
D	224----->239	1110 -----
E	240----->255	1111 -----

3- Identify the class of the following IPv4 address: 4.5.6.7.

A-A	C-C	A
B-B	D- none of the above	

4- Identify the class of the following IPv4 address: 229.1.2.3.

A-A	C-D	C
B-B	D- None of the above	

5- . Identify the class of the following IPv4 address: 191.1.2.3

A-A	C-C	B
B-B	D-None of the above	

6- What is the result of ANDing 255 and 15?

A-255	C-0	B
B-15	D- None of the above	

7- What is the result of ANDing 192 and 65?

A-192	C-64	C
B-65	D- none of the above	

8- The number of addresses in a class C block is _____

A-65,534	C-256	C
B-16,777,216	D-None of the above	

9- The number of addresses in a class B block is _____

A-65,536	C-256	A
B-16,777,216	D- none of the above	

10- The number of addresses in a class A block is _____

A- 65,534	C-256	B
B-16,777,216	D- none of the above	

11- The number of addresses assigned to an organization in classless addressing _____

A- can be any number	C- must be a power of 2	C
B- must be a multiple of 256	D- none of the above	

12-. What is the first address of a block of classless addresses if one of the addresses is 12.2.2.76/27?

A- 12.2.2.0	C-12.2.2.64	C
B-12.2.2.32	D-none	

تذكر :

CIDR : يكون بشكل IP/n

n : عدد الواحدات في mask

مثال : 8 / يكون mask هو :

11111111 00000000 00000000 00000000

← أي نضع 8 وحدات بدءاً من أقصى اليسار وباقي الخانات اصفار.

توضيح :

يوجد طريقتين لإيجاد العنوان الأول في الشبكة

الطريقة الأولى:

(1) نمثل العنوان المعطى ثنائياً :

12.2.2.76

→ 00001100 00000010 00000010 01001100

(2) نقوم بحساب $32-n$ ← $32-27=5$

(3) نحول اول 5 بت بدءا من أقصى اليمين (في العنوان الممثل ثنائيا) إلى اصفار.

→ 00001100 00000010 00000010 01000000

12 . 2 . 2 . 64

الطريقة الثانية :

نمثل العنوان المعطى والـ mask ثنائيا ثم نجري عملية And بينهما :

Address: 00001100 00000010 00000010 01001100

Mask : 11111111 11111111 11111111 11100000

First address: 00001100 00000010 00000010 01000000

12 . 2 . 2 . 64

13- . What is the first address of a block of classless addresses if one of the addresses is 12.2.2.76/10?

A- 12.0.0.0	C-12.2.2.2	A
B-12.2.0.0	D-none	

14- What is the first address of a block of classless addresses if one of the addresses is 12.2.2.127/28?

A- 12.2.2.0	C-12.2.2.112	C
B-12.2.2.96	D-none	

15- . Find the number of addresses in a block of classless addresses if one of the addresses is 12.2.2.7/24.

A- 32	C-256	C
B-64	D-none	

توضيح :

يوجد طريقتين لإيجاد عدد العناوين في الشبكة

الطريقة الأولى :

عدد العناوين في الشبكة = $2^{32-n} = 2^{32-24} = 2^8 = 256$

الطريقة الثانية :

(1) نوجد mask بما ان CIRD هو 24 /

→ 1111111 1111111 1111111 00000000

24 واحد

(2) نوجد المتمم لل mask :

(أي نقلب الازفر واحدات والواحدات افر) .

00000000 00000000 00000000 1111111

(3) نمثله عشريا ونضيف واحد

$$\rightarrow 255 + 1 = 256$$

16- Find the number of addresses in a block of classless addresses if one of the addresses is 12.2.2.7/30.

A- 2	C-8	B
B-4	D-none	

17- What is the last address of a block of classless addresses if one of the addresses is 12.2.2.127/28?

A-12.2.2.16	C-12.2.2.127	C
B-12.2.2.112	D-none	

توضيح :

يوجد طريقتين لايجاد العنوان الأخير في الشبكة :

الطريقة الأولى :

(1) نمثل العنوان المعطى ثنائيا : 12.2.2.127

→ 00001100 00000010 00000010 0111 1111

(2) نقوم بحساب $32 - n$ ← $32 - 28 = 4$

(3) نحول اول بت 4 بتات بدءا من اقصى اليمين (في العنوان الممثل ثنائيا) لـ واحدات

→ 00001100 00000010 00000010 0111 1111
12 . 2 . 2 . 127

نمثل العنوان و الـ mask ثنائيا ثم دري عملية OR بين العنوان والمتمم لـ mask :

Address :	00001100	00000010	00000010	01111111
(/28) mask :	11111111	11111111	11111111	11110000
Mask complement :	00000000	00000000	00000000	00000111
Last address :	00001100	00000010	00000010	01111111
	12	.	2	.
			2	.
				127

18- what is the last address of the block of classes addresses if one of the addresses is 12.2.2.6/30?

A-12.2.2.2	C-12.2.2.7	C
B-12.2.2.6	D- none	

19- In classless addressing, the CIDR defines the

A- net id	C- mask	C
B- host id	D-none	

20-In a block, the CIDR is /24; what is the mask?

A- 255.255.255.0	c- 255.255.0.0	A
B- 255.255.242.0	D-none	

21- In a block, the CIDR is /15; what is the mask?

A- 255.254.0.0	C- 255.255.255.0	A
B- 255.255.255.0	D-none	

22- In a block, the mask is 255.255.192.0; what is CIDR (/n)?

A- /20	C- /18	C
B- /28	D- none	

23- An IPv6 address is _____ bits long

A- 32	C- 128	C
B- 64	D- none	

24- What is the default mask for class A in CIDR notation?

A- /9	C- /16	B
B- /8	D- none	

25- What is the default mask for class B in CIDR notation?

A- /9	C- /16	C
B- /8	D- none	

26- What is the default mask for class C in CIDR notation?

A- /24	C- /16	A
B- /8	D- none	

27- An IPv6 address consists of _____ bytes.

A- 4	C- 16	C
B- 8	D- none	

28- To make addresses more readable, IPv6 specifies _____ notation

A- dotted decimal	C- both a and b	B
B- hexadecimal colon	D- none	

29- in hexadecimal colon notation, a 128-bit address is divided into _____ sections, each _____ hexadecimal digits in length

A- 8:2	C- 8:4	C
B- 8:3	D- none	

30- An IPv6 address can have up to _____ hexadecimal digits

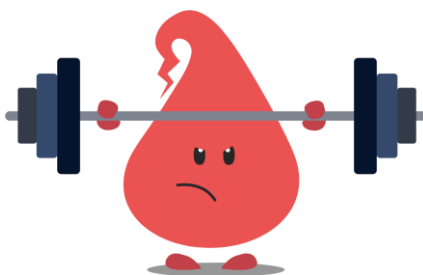
A- 16	C- 8	B
B- 32	D- none	

31- which one is not IPV4 address :

A- 111.56.045.78	C- 111.56.45.78	A
B- 111.56.450.78	D- none	

32- which one is not IPV4 address :

A- 221.43.7.8.20	C- 43.7.8.20	A
B- 221.43.7.8	D- none	



33- Identify the class of the following IPv4 address:

00000001 00001011 00001011 11101111

A-A	C-C	A
B-B	D-none	

34- Identify the class of the following IPv4 address:

11000001 00001011 00001011 11111111

A-A	C-C	C
B-B	D-none	

35- Identify the class of the following IPv4 address:

10100001 00001011 00001011 00111111

A-A	C-C	B
B-B	D-none	

36- In classful addressing , the portion of the IP address that identifies the network called the _____

A-host id	C-mask	B
B-net id	D-none	

37- In classful addressing , the portion of the IP address that identifies the host or router on the network called the _____

A-host id	C- mask	A
B-net id	D-none	

38- Addresses in the class A , B or C are mostly used for _____ communication.

A-multicast	C-unicast	C
B-broadcast	D-reserved	

39- Addresses in class D are used for _____ communication

A-multicast	C-unicast	A
B-broadcast	D-reserved	

40- Addresses in class E are _____

A-multicast	C-unicast	D
B-broadcast	D-reserved	

41- What are the restrictions in classless addressing:

A- the number of addresses needs to be a power of 2.	c- the starting address must be divisible by the number of addresses in the	D
B- the mask needs to be included in the address to define the block	D-all of the above	

42- To find the first address in a block , we set the rightmost (32-n) bits to 0

A-true	A
B-false	

43- To find the last address in a block , we set the rightmost (32-n) bits to 1

A-true

B-false

A

44- To find the number of address in the block , we calculate 2^{32-n}

A-true

B-false

A

45- There are three type of addresses in IPV6:

A-unicast ,anycast and multicast

C- broadcast , unicast and multicast

B- broadcast , anycast and multicast

D-none of the above

A

46-In classful addressing the address space is divided into _____ classes

A-6

C-5

B-7

D-14

C

47- Show the shortest form of this IPV6 address

FDEC:0074:0000:0000:0000:BOFF:0000:FFF0

A-FDEC: 74: BOFF:0: FFF0

C-FDEC:0074:0000:0000:0000:BOFF:0000:FFF0

B-FDEC: 0074:BOFF:0000:FFF0

D-None of the above

D

توضيح:

* FDEC: 0074:0000:0000:0000:BOFF:0000:FFF0

يمكننا ان نقوم ببعض الاختصارات :

(1) نختصر الازهار الموجوده على يسار كل خانه .

(2) نختصر كل 4 اصفار بصفر واحد

(3) نختصر 3 خانات المتتالية التي تساوي الصفر ونعوض عنها بمسافة 'Gap' فيصبح العنوان على هذا الشكل

→ FDEC : 74 : : BOFF: 0: FFF0

48- Show the original (unabbreviated) form of this IPV6 address 0: 15: : 1:12:1213

A-0000:0015:0000:0000:0000:0001:0012:1212	C-0000:1500:0000:0000:0000:1000:1200:1213	A
B-0000:0015:0001:0012:1213	D-none	

للتوضيح :

0: 15: : 1 : 12 : 1213

لارجاع هذا العنوان لشكله الأصلي نقوم بالتالي :

(1) يجب ان يكون في هذا العنوان 8 خانات وكل خانة عبارة عن 4 ارقام ، فنقوم بارجاع الاصفار التي تم اختصارها من يسار كل خانة

(2) نقوم بارجاع 3 خانات متتالية من الاصفار التي تم اختصارها بمسافة .

X X X X : X X X X : X X X X : X X X X : X X X X : X X X X : X X X X : X X X X
0 : 15 : : : : 1 : 12 : 1213

→ original address is :

0000 : 0015 ; 0000 :0000: 0000 : 0001 : 0012 : 1213



Internet Protocol

1) Communication at the **network layer** in the Internet is:

A- Connectionless	C- Both are	A
B- Connection oriented	D- None of the above	

2) Packets in the IPV4 layer are called:

A- Frames	C- Datagrams	C
B- Segments	D- Signals	

3) In which layer is the IPV4 protocol located?

A- Application layer	C- Datalink layer	B
B- Network layer	D- None of the above	

4) The UDP , TCP and SCTP protocols are located in:

A- Application layer	C- Datalink layer	D
B- Network layer	D- Transport layer	

5) The SMTP, FTP, TFTP , DNS ,SNMP and BOOTP protocols are located in:

A- Application layer	C- Datalink layer	A
B- Network layer	D- Transport layer	

6) The ARP, RARP, ICMP and IGMP protocols are located in:

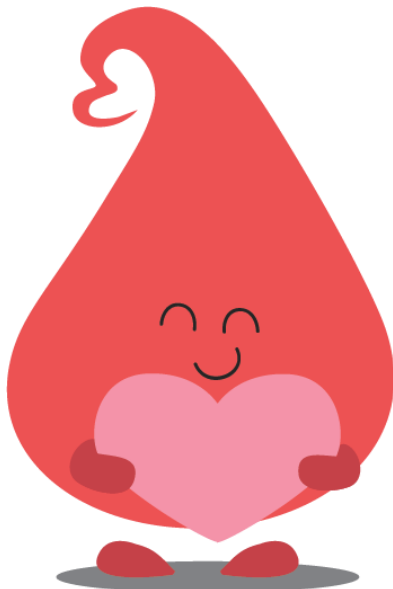
A- Application layer	C- Datalink layer	B
B- Network layer	D- Transport layer	

7) IP is _____ datagram protocol:

A- an unreliable	C- both a and b	C
B- a connectionless	D- none of the above	

8) A router reads the _____ address on a packet to determine the next hope:

A- MAC	C- IP	C
B- source	D- ARP	



وتهدينا الحياة أضواء في آخر النفق

تدعونا كي ننسى ألماً عشناه

نستسلم لكن لا ما دمنا أحياء نرزق

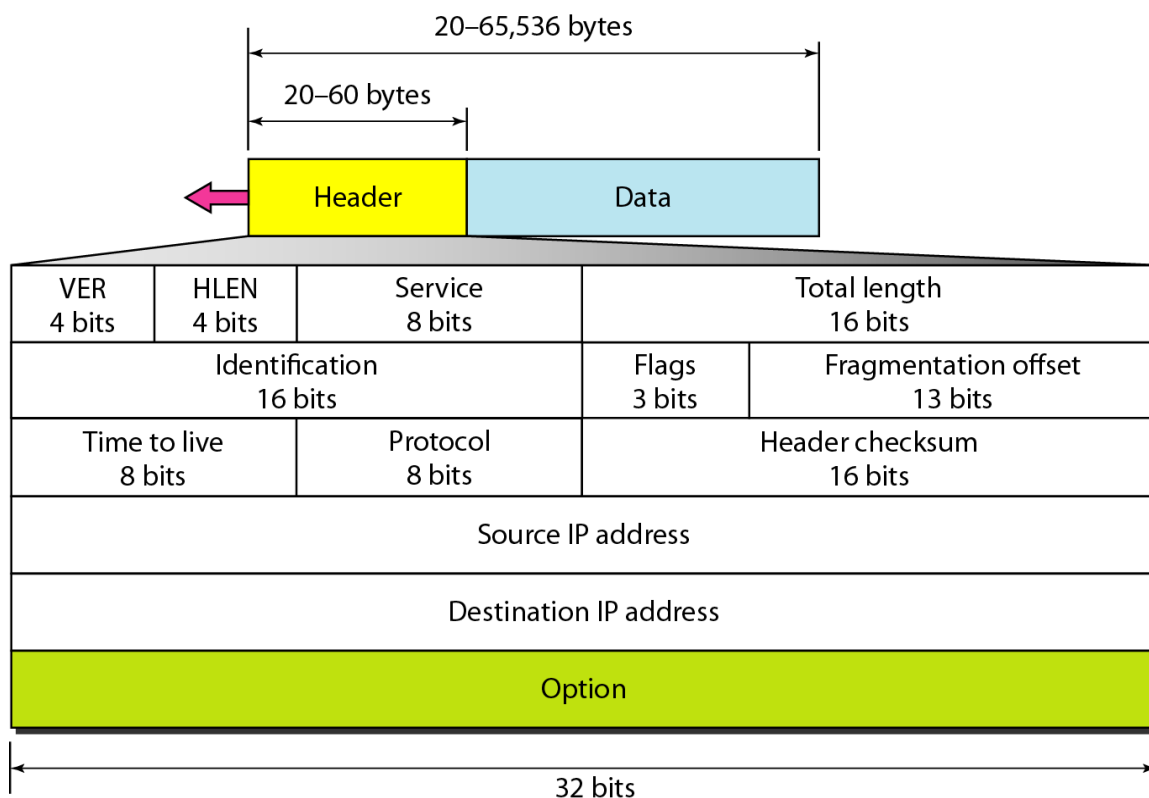
ما دام الأمل طريقاً فسنحياه

9) IPV4 is **unreliable connectionless** protocol responsible for source to destination delivery:

A- True	B- False	A

ملاحظة: يجب حفظ هذه الرسمة ومعرفة موقع كل حقل بالترتيب ومن كم بت يتألف.

IPv4 datagram format



10) A datagram consists of a _____ :

A- header (20 to 60 bytes) and data and trailer	C- data and trailer	B
B- header (20 to 60 bytes) and data	D- none of the above	

11) The IPv4 header size _____ :

A- is 20 to 60 bytes long

C- is always 60 bytes long

B- is always 20 bytes long

D- depends on the MTU

A

12) The maximum length of a datagram _____ :

A- is 20 bytes

C- is 60 bytes

B- 65,535 bytes

D- depends on the MTU

B

13) In IPv4, when a datagram is encapsulated in a frame, the total size of the datagram must be less than the _____ :

A- MUT

C- MTU (Maximum Transmission Unit)

B- MAT

D- none of the above

C

14) In IPv4, what is the value of the total length field in bytes if the header is 28 bytes and the data field is 400 bytes?

A- 428

C- 107

B- 407

D- 427

A

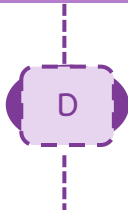
توضيح:

لحساب قيمة الحقل total length: نجمع طول الهيدر مع طول البيانات القادمة من الطبقة العليا فيكون:

Total length = length of data + header length

Total length = 400 + 28 = 428

15) In IPv4, what is the length of the data field given an HLEN value of 12 and total length value of 40,000?

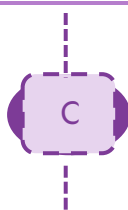
A- 39,988	C- 40,048	
B- 40,012	D- 39,952	

توضيح:

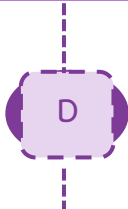
Length of data = total length – header length

Length of data = $40000 - (12 \times 4) = 39952$

16) An IPv4 datagram is fragmented into three smaller datagrams. Which of the following is true?

A- The do not fragment bit is set to 1 for all three datagrams.	C- The identification field is the same for all three datagrams.	
B- The more fragment bit is set to 0 for all three datagrams	D- The offset field is the same for all three datagrams	

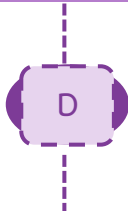
17) In IPv4, if the fragment offset has a value of 50, it means that _____ :

A- the datagram has not been fragmented	C- the first byte of the datagram is byte 100	
B- the datagram is 100 bytes in size	D- the first byte of the datagram is byte 400	

توضيح:

First byte = offset $\times$ 8 = $50 \times 8 = 400$

18) In IPv4, what is needed to determine the number of the last byte of a fragment?

A- Identification number	C- Total length	
B- Offset number	D- (b) and (c)	

19) Which of the following is a necessary part of the IPv6 datagram?

A- Base header	C- Data packet from the upper layer	A
B- Extension header	D- (a) and (c)	

20) In IPv4, an HLEN value of decimal 10 means _____ :

A- there are 10 bytes of options	C- there are 40 bytes in the header	C
B- there are 40 bytes of options	D- none of the above	

21) The IPv4 header field formerly known as the service type field is now called the _____ field:

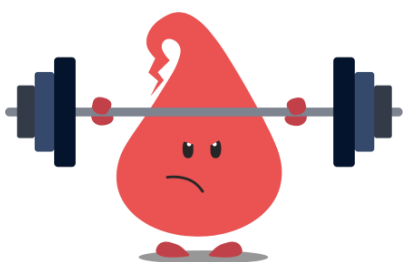
A- IETF	C- differentiated services	C
B- checksum	D- none of the above	

22) In IPv6, options are inserted between the _____ and the _____ data:

A- base header, extension header	C- base header, frame header	B
B- base header, upper-layer data	D- none of the above	

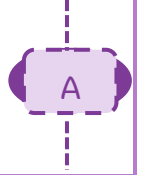
23) IPv6 allows _____ security provisions than IPv4:

A- more	C- the same level	A
B- less	D- none of the above	



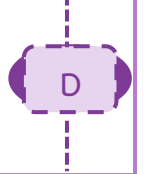
24) The total length field defines _____ :

- | | |
|--|---|
| A- the total length of the datagram including the header | C- the total length of the datagram including the options |
| B- the total length of the datagram | D- none of the above |



25) An IPv4 packet has arrived with the first 8 bits as shown: 01000010

- | | |
|---|---|
| A- The receiver will discard the packet | C- The receiver will not discard the packet |
| B- There is an error in this packet | D- A and B |



توضيح:

يوجد خطأ في هذه الحزمة 0100 0010

- حيث الـ 4 بت على اليسار (0100) تمثل الإصدار (IPv4) وهذا صحيح لأن 4 = 0100

- الـ 4 بت التالية (0010) تمثل طول الهيدر (HLEN) و 2 = 0010

$$\Rightarrow 2 \times 4 = 8 \text{ bytes}$$

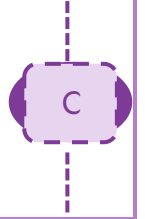
وهذا يظهر أن Header length غير صالح لأن الحد الأدنى لعدد البايتات في هيدر الـ IPv4 يجب أن يكون

20 bytes

وبالتالي يتم رفض هذه الحزمة.

26) In an IPv4 packet, the value of HLEN is 1000 in binary. How many bytes of options are being carried by this packet?

- | | |
|-------------|----------------------|
| A- 13 bytes | C- 12 bytes |
| B- 14 bytes | D- none of the above |



توضيح:

إذا كان header length يساوي 1000 ثنائياً و يساوي 8 عشرياً مما يعني أن إجمالي عدد البايتات في الهيدر هو:

$$8 \times 4 = 32 \text{ bytes}$$

وكما نعلم أنه يجب أن يكون عدد البايتات في الهيدر كحد أدنى 20 bytes وتكون للمعلومات الأساسية والباقي من

البايتات ستكون للخيارات الإضافية

$$\Rightarrow 32 - 20 = 12$$

27) The value of HLEN in an IPV4 datagram is 7 . How many option bytes are present?

A- 8 bytes

C- 12 bytes

B- 14 bytes

D- none of the above

A

28) In an IPv4 packet, the value of HLEN is 5, and the value of the total length field is 0x0028. How many bytes of data are being carried by this packet?

A- 20 bytes

C- 80 bytes

B- 40 bytes

D- 64 bytes

A

توضيح:

إذا كان طول الهيدر يساوي 5 هذا يعني أن إجمالي عدد البايتات في الهيدر هو $5 \times 4 = 20$ bytes

وإذا كان الطول الكلي للحزمة هو 0x0028 بالسداسي عشر ويساوي 40 عشرياً

هذا يعني أن الحزمة تحمل 20 بايت من البيانات وذلك حسب:

$$\text{Length of data} = \text{total length} - \text{header length}$$

$$\text{Length of data} = 40 - 20 = 20 \text{ bytes}$$

29) An IPv4 packet has arrived with the first few hexadecimal digits as shown:

0x45000028000100000102

How many hops can this packet travel before being dropped? The data belong to what upper-layer protocol?

A- One hop, IGMP

C- two hops, IGMP

B- one hop, TCP

D- Two hops, TCP

A

توضيح:

يمكننا تحديد حقل Time to live بتجاوز 8 bytes من البيانات (أي يمثل هذا الحقل بالبايت التاسع) والذي يكون بقيمة

01 وهذه القيمة تشير أن الحزمة يمكنها القفز عبر قفزة واحدة فقط.

والبايت التالي يمثل حقل البروتوكول وقيمته 02 والذي يعني أن البروتوكول هو IGMP

تم معرفة البروتوكول من هذا الجدول:

Value	Protocol
1	ICMP
2	IGMP
6	TCP
17	UDP
89	OSPF

30) The value of the header length in an IPV4 packet can be less than 5:

A- True	B- False	B

31) A packet has arrived with an M bit value of 0. we can't know if the packet was fragmented:

A- True	B- False	A

توضيح:

وصلت الحزمة وكانت $M = 0$ (M ترمز للبت الثالث في حقل (Flags (more fragment)). إذاً إما الحزمة أصلية أي ليست مجزأة أو مجزأة وهذه الحزمة هي الجزء الأخير.

32) A packet has arrived with an M bit value of 1. we can't know if the packet was fragmented:

A- True	B- False	B

توضيح:

وصلت الحزمة وكانت $M = 1$ إذاً الحزمة مجزأة وهذه الحزمة ممكن أن تكون الجزء الأول أو واحدة وسطى، ولكنها ليس الجزء الأخير.

33) A packet has arrived with an M bit value of 1 and a fragmentation offset value of 0:

A- It is the first fragment	C- It is the middle fragment	
B- It is the last fragment	D- A and C	

توضيح:

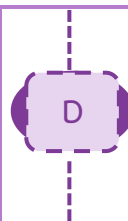
وصلت الحزمة وكان $M = 1$ و $offset = 0$ إذاً الحزمة مجزأة إلى حزم صغيرة، وهذه الحزمة إما تكون الجزء الأول أو واحدة وسطى، وبما أن $offset = 0$ فإن الحزمة هي الجزء الأول.

ملاحظة: لو كان $M = 0$ و $offset = 0$ فستكون هذه الحزمة أصلية غير مجزأة.

34) A packet has arrived in which the offset value is 100 What is the number of the first byte?

A- 800	C- 400	
B- 100	D- Cannot determine	

35) A packet has arrived the offset value is 100. What is the number of the last byte?

A- 800	C- 400	
B- 100	D- Cannot determine	

36) A packet has arrived in which the offset value is 100, the value of HLEN is 5, and the value of the total length field is 100. What are the numbers of the first byte and the last byte?

A- first bytes = 800, last byte = 879	C- first bytes = 400, last byte = 879	
B- first bytes = 800, last byte = 877	D- Cannot determine	

توضيح:

وصلت حزمة $offset = 100$ و $HLEN = 5$ فيكون طول الهيدر هو $5 \times 4 = 20$ و $total\ length = 100$
- لحساب رقم البايت الأول:

$$\text{First byte} = \text{offset} \times 8 = 100 \times 8 = 800 \text{ bytes}$$

- لحساب رقم البايت الأخير:

$$\text{Length of data} = \text{total length} - \text{header length} = 100 - 20 = 80$$

← إذا كان البايت الأول هو 800 فالبايت الأخير هو 879.

37) What is needed to determine the number of the last byte of a fragment ?

A- Identification	C- Total length	D
B- Offset number	D- B and C	

38) In IPv6, the _____ field in the base header restricts the lifetime of a datagram:

A- version	C- hop limit	C
B- next-header	D- neighbor-advertisement	

39) _____ is the division of a datagram into smaller units to accommodate the MTU of a data link protocol:

A- Fragmentation	C- Offset	A
B- Identification	D- none of the above	

40) The _____ section of the IPV4 header is used for network testing and debugging:

A- options	C- flags	A
B- time to live	D- service	

41) Three strategies used to handle the transition from version 4 to version 6 are:

A- single stack , tunneling and header translation	C- dual stack , tunneling and option translation	B
B- dual stack , tunneling and header translation	D- none of the above	

42) _____ is a strategy used when two computers using IPV6 want to communicate with each other and the packet must pass through a region that uses IPV4:

A- Tunneling

C- Header translation

B- Dual stack

D- none of the above

A

43) _____ is a necessary when the majority of the Internet has moved to IPV6 but some system still use IPV4 , the sender wants to use IPV6 but the receiver doesn't understand IPV6:

A- Tunneling

C- Header translation

B- Dual stack

D- none of the above

C

44) The strategies when a station must run IPV4 and IPV6 simultaneously until all the Internet uses IPV6 is:

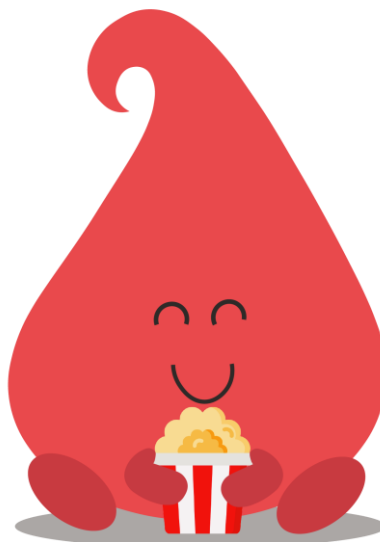
A- Tunneling

C- Header translation

B- Dual stack

D- none of the above

B





Address Mapping and Error Reporting

1. The **logical addresses** in the Internet are called _____ addresses

A. port

C. Email

B. IP

D. none of the above

B

2. If the **sender is a host** and wants to send a packet to another **host on the same network**, the logical address that must be mapped to a physical address is _____

A. the destination IP address in the datagram header

C. either a or b

B. the IP address of the router found in the routing table

D. none of the above

A

3. If the **sender is a host** and wants to send a packet to another **host on another network**, the logical address that must be mapped to a physical address is _____.

A. the destination IP address in the datagram header

C. either a or b

B. the IP address of the router found in the routing table

D. none of the above

B

4. The **sender is a router** that has received a datagram destined for a **host on another network**. The logical address that must be mapped to a physical address is _____.

A. the destination IP address in the datagram header

C. either a or b

B. the IP address of the router found in the routing table

D. none of the above

B

5. The sender is a router that has received a datagram destined for a host on the same network. The logical address that must be mapped to a physical address is _____.

- | | |
|--|----------------------|
| A. the destination IP address in the datagram header | C. either a or b |
| B. the IP address of the router found in the routing table | D. none of the above |

A

6. In _____, a table associating a logical address with a physical address is updated manually.

- | | |
|--------------------|----------------------|
| A. static mapping | C. physical mapping |
| B. dynamic mapping | D. none of the above |

A

7. _____ is a dynamic mapping protocol in which a physical address is found for a given logical address.

- | | |
|---------|----------------------|
| A. ARP | C. both a and b |
| B. RARP | D. none of the above |

A

8. The target hardware address on an Ethernet is _____ in an ARP request

- | | |
|-------------------|--------------------|
| A. 0x000000000000 | C. variable |
| B. 0.0.0.0 | D. class dependent |

A

9. An ARP reply is normally _____

- | | |
|--------------|----------------------|
| A. broadcast | C. unicast |
| B. multicast | D. none of the above |

C

10. An ARP request is normally _____

A. broadcast

C. unicast

B. multicast

D. none of the above

A

11. A technique called _____ is used to create a subnetting effect.

A. ARP

C. proxy ARP

B. RARP

D. none of the above

C

12. A _____ is an ARP that acts on behalf of a set of hosts

A. ARP

C. proxy ARP

B. RARP

D. none of the above

C

13. ICMP is a _____ layer protocol

A. data link

C. network

B. transport

D. none of the above

C

14. ICMP messages are divided into two broad categories:

A. query and error reporting messages

C. request and reply messages

B. request and response messages

D. none of the above

A

15. An ICMP message has _____ header and a variable-size data section .

A. a 16-byte

C. an 8-byte

B. a 32-byte

D. none of the above

C

16. Which of the following is true about ICMP messages?

- | | |
|---|---|
| A. An ICMP error message may be generated for an ICMP error message | C. An ICMP error message may be generated for a multicast datagram. |
| B. An ICMP error message may be generated for each fragment | D. none is true |

D

17. Which of the following is true about ICMP messages?

- | | |
|---|---|
| A. An ICMP error message may be generated for an ICMP error message | C. An ICMP error message may be generated for a multicast datagram. |
| B. An ICMP error message may be generated only for the first fragment | D. none is true |

B

18. A multicast message is sent from _____ to _____.

- | | |
|--------------------------------------|--------------------------------------|
| A. one source; one destination | C. multiple sources; one destination |
| B. one source; multiple destinations | D. none of the above |

B

19. Delivery of a packet to a host or router requires :

- | | |
|---|---|
| A. two levels of addresses are logical and physical . | C. one level of addresses is physical . |
| B. one level of addresses is logical . | D. none of above |

A

20. Mapping of a logical address to a physical address can be :

- | | |
|------------------------|------------------|
| A. static and dynamic. | C. only static. |
| B. static or dynamic. | D. only dynamic. |

B

21. In , a router represents a set of hosts . when an ARP request seeks the physical address of any host in this set , the router sends its own physical address

A. RARP	C. ICMP	B
B. proxy ARP	D. none of the above	

22. sends pairs of query messages : (echo, time stamp , address _ mask) request and reply.

A. RARP	C. ICMP	C
B. proxy ARP	D. none of the above	

23. The checksum for ICMP is calculated by using :

A. both the header and the data fields of the ICMP message .	C. the data fields of the ICMP message.	A
B. the header and the data fields of the ICMP message and trailer.	D. none of the above	

24. is a router that helps send local BOOTP request to remote servers.

A. proxy ARP	C. BOOTP client	B
B. rely agent	D. BOOTP server	

25. The server issues a lease for an IP address to a client for a **specific time**

A. DHCP	C. BOOTP	A
B. ARP	D. ICMP	

26. provides static and dynamic address allocation that can be manual or automatic.

A. DHCP

C. BOOTP

B. ARP

D. ICMP

A

27..... always reports error messages to the original source

A. DHCP

C. BOOTP

B. ARP

D. ICMP

D

28. The has no error-reporting or error-correcting mechanism

A. DHCP

C. BOOTP

B. IP protocol

D. ICMP

B

ملاحظة:

الحالات التي لا يجب فيها إنشاء رسائل خطأ في بروتوكول ICMP :

عدم إنشاء رسالة خطأ ICMP رداً على حزمة بيانات تحمل رسالة خطأ ICMP وذلك لتجنب دورة لا نهائية لها من رسائل الأخطاء.

عدم إنشاء رسالة خطأ ICMP لحزمة بيانات مجزأة ليست first fragment.

عدم إنشاء رسالة خطأ ICMP لحزمة بيانات تحمل عنواناً نوعه Multicast.

عدم إنشاء رسالة خطأ ICMP لحزمة بيانات تحمل عنواناً خاصاً مثل : 127.0.0.0 أو 0.0.0.0

29. No ICMP error message will be generated in response to a datagram carrying an ICMP error message.

A. true

B. false

A

30. No ICMP error message will be generated for a fragmented datagram that is not the first fragment

A. true

B. false

A

31. No ICMP error message will be generated for a datagram having a multicast address.

A. true

B. false

A

32. No ICMP error message will be generated for a datagram having a special address such as 127.0.0.0 or 0.0.0.0

A. true

B. false

A

على أمل أن تتحقق الأحلام التي نبذل فيها يومنا ،
ونومنا وشرًا من أرواحنا ، ونحيا بقية العمر وما سعيها له بين أيدينا.

لا تنسونا من دعواتكم والسلام عليكم ♥

